

Maryam Amjad

Machine Learning / AI Engineer | LLM & RAG | Deep Learning | Computer Vision
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SUMMARY

AI/ML Engineer with about a year of professional experience across crypto, data-intelligence, and healthcare AI. Built end-to-end systems: a real-time crypto prediction pipeline, a RAG-based text-to-SQL engine using knowledge-graph and vector retrieval, a Neo4j graph analysis app, and a patient-facing multi-modal AI health assistant, plus a lead generation and automated outreach tool. Core strength: taking a system from raw data to something people actually use.

EXPERIENCE

Junior Machine Learning Engineer

Tensor Labs

Sep 2025 – Aug 2026

Worked on production ML/LLM systems across crypto, data-intelligence, and healthcare products, including data pipelines, model integration, and reliability.

Real-Time Crypto Breakout Prediction System

Problem: Traders needed early breakout signals across multiple volatile crypto markets from continuous streaming data.

Role: Designed and deployed the real-time prediction pipeline end-to-end.

Tech: Python, Pandas, Scikit-learn, PostgreSQL, WebSockets, Streamlit.

Outcome: Live system processing 5-minute streaming data for ETH, DOGE, and SOL, serving breakout alerts and a liquidation/liquidity-risk heatmap through a real-time dashboard.

RAG-Based Text-to-SQL Engine (DiscoverIQ)

Problem: Non-technical users needed to query large relational databases without writing SQL, but LLM-generated SQL often failed on real schemas on the first try.

Role: Built the self-correcting SQL feedback loop: when a generated query fails, the agent analyzes the error, asks the model for a corrected query, validates syntax before re-running it, and retries with duplicate-query detection.

Tech: Python, OpenAI, Groq (Llama-3.3-70B), Neo4j, Qdrant, PostgreSQL, JSON repair/validation.

Outcome: Automatically recovers from failed SQL generations without user intervention, using error-analysis prompting, syntax pre-validation, and duplicate-query detection to converge on a working query.

MedCompanion: Patient-Facing AI Health Assistant (MedGemma & TxGemma)

Problem: Patients hold fragmented records, prescriptions they cannot interpret, and no easy way to spot dangerous drug interactions.

Role: Part of a 3-person team. Built the prescription and medicine OCR module, the drug interaction checker, and the vitals, symptoms, allergy, and medicine-intake tracking.

Tech: MedGemma-4B, TxGemma-9B, Qwen3-VL-8B, Whisper-large-v3, Streamlit, Docker, FDA/RxNorm/Wikipedia APIs.

Outcome: Working assistant that analyzes medical scans, reads prescriptions, checks drug interactions, transcribes clinical audio, and answers questions from the patient's own health history, with PDF reports.

Straight Egyptian Horse Graph Analysis (Neo4j)

Problem: Data on horse lineages, stables, ownership, and competition results was interconnected, and spreadsheets made it hard to see patterns or choose good breeding pairs.

Role: Converted raw Excel data into a Neo4j graph of horses, stables, breeders, and competitions, and built the Streamlit and PyVis interface and the breeding-pair scoring engine.

Tech: Python, Pandas, Neo4j, Streamlit, PyVis, Docker.

Outcome: Interactive app for exploring stables, 5-generation pedigrees, and top-10 horses per performance metric, plus a ranked table of recommended breeding pairs scored on individual and parent rankings, stable performance, and competition wins.

Jun 2023 – Aug 2023

Built a tool that auto-generates presentation slide decks from natural-language prompts using NLP and computer vision; applied pre-trained vision models for content understanding and image matching.

PROJECTS

Lead Finder and Automated Outreach Tool: Finds business leads for a given service and location using OpenStreetMap listings plus free web search, filters out directories and irrelevant sites, scrapes contact details from company websites, and assigns a confidence level to each result. Stores results in a running database and sends personalized outreach emails using a separate template per service, with rate limiting and a suppression list so no contact is messaged twice. Working and in active use. *Python, Streamlit, OpenStreetMap API, DuckDuckGo search, web scraping, email automation.*

LLM Evaluation Toolkit: Open-source toolkit covering 12 LLM evaluation categories (text generation, summarization, RAG, text-to-SQL, safety & robustness, long-context reasoning, multi-modal) with runnable notebooks and an interactive Streamlit dashboard; used as the basis for an internal engineering workshop. *Python, Groq, RAGAS, DeepEval, HuggingFace Evaluate, PromptBench.*

TherapEase (Autism Therapy Assistant): 3D digital twin with real-time facial emotion detection to support autism therapy sessions, combining a browser-based 3D avatar with a live computer-vision emotion-classification backend (final year project). *React, Three.js, OpenCV, MediaPipe, Flask, TensorFlow, DeepFace.*

Roman Urdu Chatbot: Rasa-based chatbot for Roman Urdu small talk with 30+ intents and custom Python actions, plus a voice-to-voice mode using Google Speech-to-Text and gTTS. Deployed on Streamlit Cloud. *Rasa, Streamlit, gTTS, SpeechRecognition, Python.*

SKILLS

ML/AI: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, RAG, Prompt Engineering, Agentic Workflows, RAGAS, DeepEval

Domains: Computer Vision, NLP, LLMs/Agents, Knowledge Graphs, Time Series, OCR/Speech (Whisper, Mem0)

Data: Neo4j, Qdrant, PostgreSQL, MySQL, SQLAlchemy, Pandas, NumPy, OpenCV, NLTK, SpaCy, Web Scraping

Cloud/DevOps: AWS, Docker, FastAPI, Flask, Streamlit, CI/CD, Git

EDUCATION

B.S. in Artificial Intelligence

Coursework: Deep Learning, Generative AI, Computer Vision, NLP

FAST NUCES, Islamabad

Jul 2021 – Jun 2025

CERTIFICATIONS

- AWS Certified AI Practitioner (2026)
- Generative AI with LLMs, DeepLearning.AI (2025)
- AI Agents with RAG and LangChain, IBM (2025)
- Convolutional Neural Networks, DeepLearning.AI (2024)
- AWS Cloud Essentials (2025)